

Asu's Guide to the ACE WORKSHOP

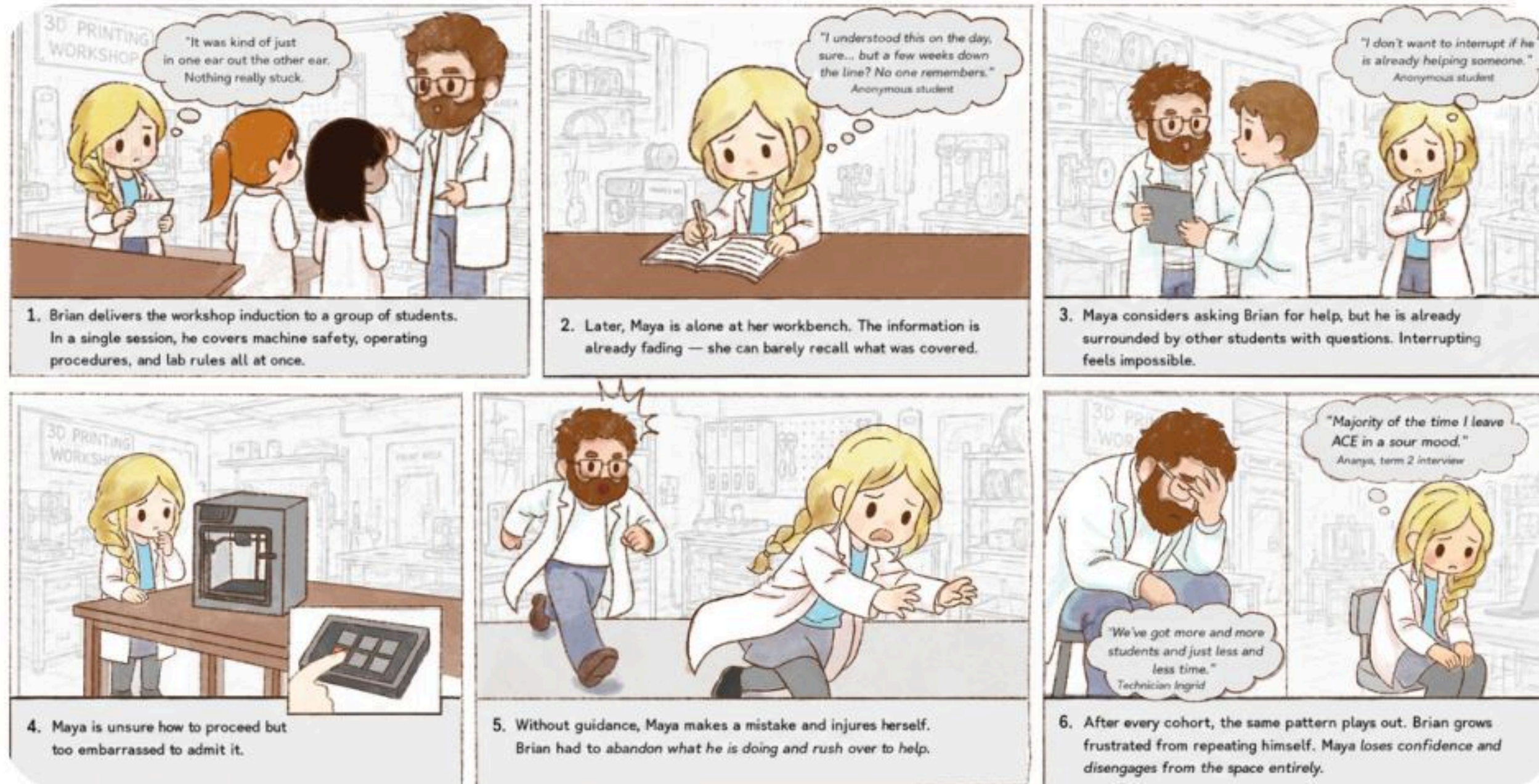
 **Learn it** ✨
 **Practice it**
 **Print it!** ✨

Asteria Wang
Sarah Xie
Elena Zhuang
Joseph Zhang

Executive Summary

Current user journey

See Appendix for the full development process of the storyboard



Stakeholder Persona

Primary users: Students who are new to the Ace Workshop



Need: A clear, low-pressure entry point that **builds confidence** before entering the workshop.

Pain point: Without prior context, the volume of induction information feels **overwhelming**, leading to **passive listening** and **poor retention**.

Secondary users: Students who need a refresher using machines in ACE



Need: Quick, targeted access to specific rules or procedures **without repeating** the full induction.

Pain point: No recall mechanism exists, forcing students to either **interrupt** technicians or **proceed uncertainly**.

Tertiary stakeholders: Workshop technicians and workshop manager



Need: Students to arrive **prepared** and **self-sufficient**, reducing **repetitive** low-level queries.

Pain point: Constant **re-explanation** of basic induction content during peak periods **drains time** and **disrupts high-value technical support**.

Survey: 48% dissatisfied

48% of survey respondents rated the current ACE workshop experience as somewhat to very **dissatisfactory**.

20 hours of observation

20 hours of ACE observation: Students were seen **scrolling**, **fidgeting** or **disengaging** during induction

18 users interviewed

18 users interviewed across **student** and **technician** perspectives

During induction...



Core insights derived

- Students **retain very little** from a **single dense induction** session
- The **social cost** of asking for help and making mistakes is as **prohibitive** as the technical problem itself
- The **informal help system** structurally penalises the most **vulnerable students**

See Google Drive: "Photographs, Recordings and Transcripts --> Photographs --> for induction observation 8th May"

Design challenge

How might we make ACE induction memorable and revisitable, so students can use equipment confidently without relying on informal help?

Solution statement

A gamified 3D printing simulator that **reinforces ACE workshop induction** through **on-demand, self-directed practice** — building student **confidence** and **reducing reliance** on technician support.

Core value

Learn it. Practice it. Print it.

Initial concepts



A series of **workshop-themed blind-box figures**. Every figure includes an ID card and an **embedded NFC** chip linking to an **induction game**.



An **electric candle** that helps students **manage focus** and **avoid burnout**.



A **fighting game** that turns inductions into timed machine battles, helping students **learn workshop rules** through **play, practice**, and **progression**.

SELECTED: see next page

From Insight to Design Requirements

DATA FROM PREVIOUS TERM + NEW LAYER OF ANALYSIS (REQUIREMENTS)

Evidence logic map

Research evidence is grouped first, then translated into opportunities, constraints, and final requirements.

1. Insights (I)

“It was kind of just in one ear out the other ear.” – P24

“I don't know who's available, what they can help.” – P25

One-off induction I.1
Students **forget dense content**.

Hidden knowledge I.2
Technician **expertise is not visible at point of need**.

“I'm afraid it might be a stupid question.” – P21

“Whoever is most assertive make people go to them.” – P9

Fear of judgement I.3
Basic questions feel **socially costly**.

Informal help I.4
Assertive students get support first.

“I have to ask every time if I'm gonna use for the first time.” – P1

“I can't remember everything at once, they rush you out.” – P21

High help cost I.5
Asking can feel as **hard** as the **technical task**.

Incomplete info I.6
No shared touchpoint gives everyone a **full picture**.

2. Design Opportunities (DO)

DO.1
Repeatable knowledge
Self-serve, machine-contextual support after the live session.
I.1 I.2 I.6

DO.2
Low-pressure learning
Game-based practice that normalises mistakes.
Sources: I3, I4, I5.
I.3 I.4 I.5

3. Constraints (C)

C.1
Busy workshop
Must work outside peak pressure times.

C.2
Formal induction
Must complement the existing system.

C.3
Staff workload
Must not need extra technician input.

4. Criteria generated (CG)

CG.1
Effective learning
The game shall demonstrably **improve knowledge retention** of induction content compared to the current single-session format
DO.1 DO.2

CG.2
On demand access
The game shall deliver content **contextualised to specific machines, accessible** on-demand **outside of peak hours**.
DO.1 C.1

CG.3
Independent confidence
The game shall operate fully **autonomously** without requiring technician facilitation or input at any stage.
DO.1 DO.2 C.3

CG.4
Engagement
The game shall be **engaging** by incorporating a **no-penalty mistake mechanism** with **immediate feedback** to build student **confidence**.
DO.2

CG.5
Induction Complement
The game shall function as a **reinforcement tool** that **complements** but does not replace the compulsory ACE **safety induction**.
DO.1 C.2

Why did we select the concept of gamified learning tool?

We selected a **gamified learning tool** as our concept direction because it **directly addressed** our **defined criteria** – and because evidence supports it.

Research across **365 engineering and business students** found that **challenge-based gamification significantly improved learning outcomes**, with the **strongest effect** seen in **engineering students** specifically – closely mirroring the **ACE student demographic**. [2]

NEW DATA FROM THIS TERM

Co-Design: Defining Game Mechanics Through User Preference

Aim: To identify **which game mechanics resonate with ACE workshop users** before concept development.

Participants: 4 DE students

Demographics: a range from **workshop beginners** to **experienced makers** owning 3D printers at home.

Session: ~45 min

Method: We listed **2 assumptions** to test beforehand (see below). Users designed their **own induction game** using **mechanism** and **constraint cards** while thinking out loud.

Participants' Quotes

Wants
P21: “**Live feedback** would have solved 90% of the problems.”
P1: “It should be something **actionable**... not just multiple-choice answers.”
P23: “**Revisitable... always-accessible** reference guides.”

Don't wants
P1: “**Timer is not necessary**... it encourages **cheating** rather than learning.”
P21: “Don't tell me about **memorising**... we will forget.”

Card Design

We designed the cards to provide concrete prompts that help participants articulate preferences they might otherwise struggle to express unprompted.



See Google Drive: “Photographs, Recordings and Transcripts --> Photographs --> for co-design session 6th May”

Codesign Session



Assumption → Outcome

A1: **Time pressure** may increase **engagement**.

→ Outcome: Users preferred **no time pressure** because it encouraged **guessing/cheating**.

A2: Rewards motivate completion.

→ Outcome: Rewards were **repeatedly selected**; **replayability** and **feedback** were also valued.



Final output from this page

Therefore, the design requirements define a **self-paced, revisitable gamified learning tool** that improves **induction knowledge retention** through **low-pressure interaction**, **reward-based engagement**, and **on-demand access** to machine-specific guidance.

Game Prototyping Journey

We developed three low-fi prototypes — rough, divergent concepts **designed to provoke reaction** rather than present a finished solution. Each **exaggerated a different game format**: dress-up quiz, RPG quest, and 3D printing simulator, helping us test **which assumptions resonated most with users**. Testing involved 13 non-Design Engineering students with no 3D printing experience **to assess clarity and learning effectiveness**, and 4 Design Engineering students with prior exposure **to assess desirability**, as their tastes were closer to incoming freshers. We also ran **an internal team vote** as a **parallel AB test** to capture our collective preference.

Concept 01

Dress-up quiz game

A **reward-led quiz concept** where safety/workflow questions **unlock coins, avatar customisation, and a printable workshop ID or badge**.



User answer safety/workflow questions to **earn coins**.



After the quiz, an **error log** lets users review mistakes.



Coins can be spent on avatar **clothes and props**.

What we tested

See Appendix for the development process

Whether **rewards, coins** and **customisation** would make quiz-based induction **more motivating** than a standard written quiz.

P8: I liked the **deduction of the points**, if you get it **wrong the first time**, encourages you to **learn** and then **remember it**... it feels like you're **working towards a goal**.

P8: But I don't like how much it is like a **quiz**.

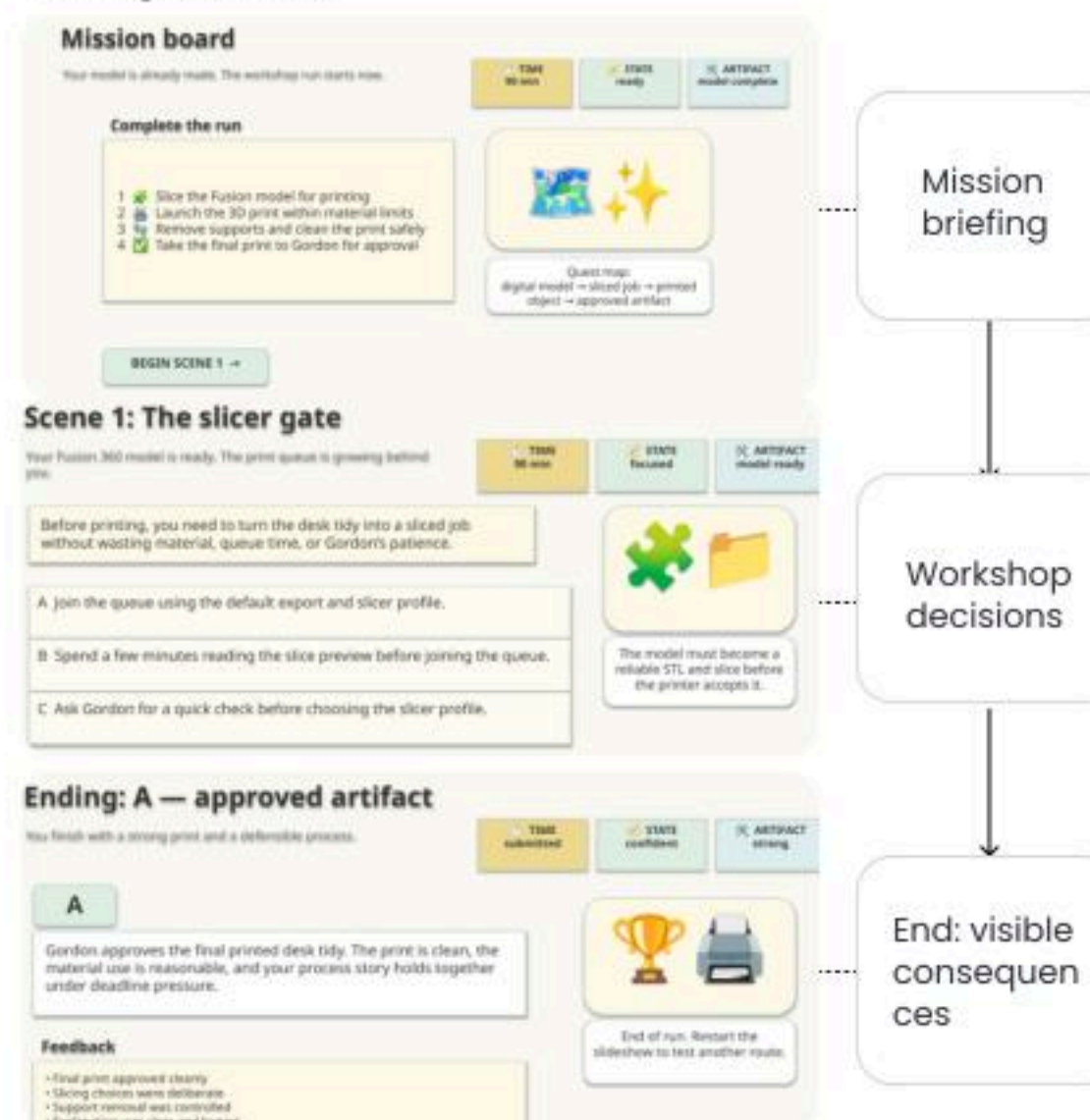
Reward loop **raised motivation**, but the learning still **depended too much on recall** and weak workshop transfer.

Decision: retain rewards as a motivator, but not as the main learning format.

Concept 02

Workshop quest: desk-tidy run

A **story-based RPG prototype** that frames workshop rules as **missions, choices, and visible consequences** in a playful making scenario.



What we tested

See Google Drive: "Prototypes"

Whether **missions, scenes** and **endings** could make workshop rules **feel less intimidating** and **easier to remember**.

P8: I think it's a bit **too centred towards our coursework** rather than the induction itself...

P8: It felt too much like a **gamified version** of what's on Blackboard.

Narrative improved engagement, but users still needed **clearer links to real workshop decisions**.

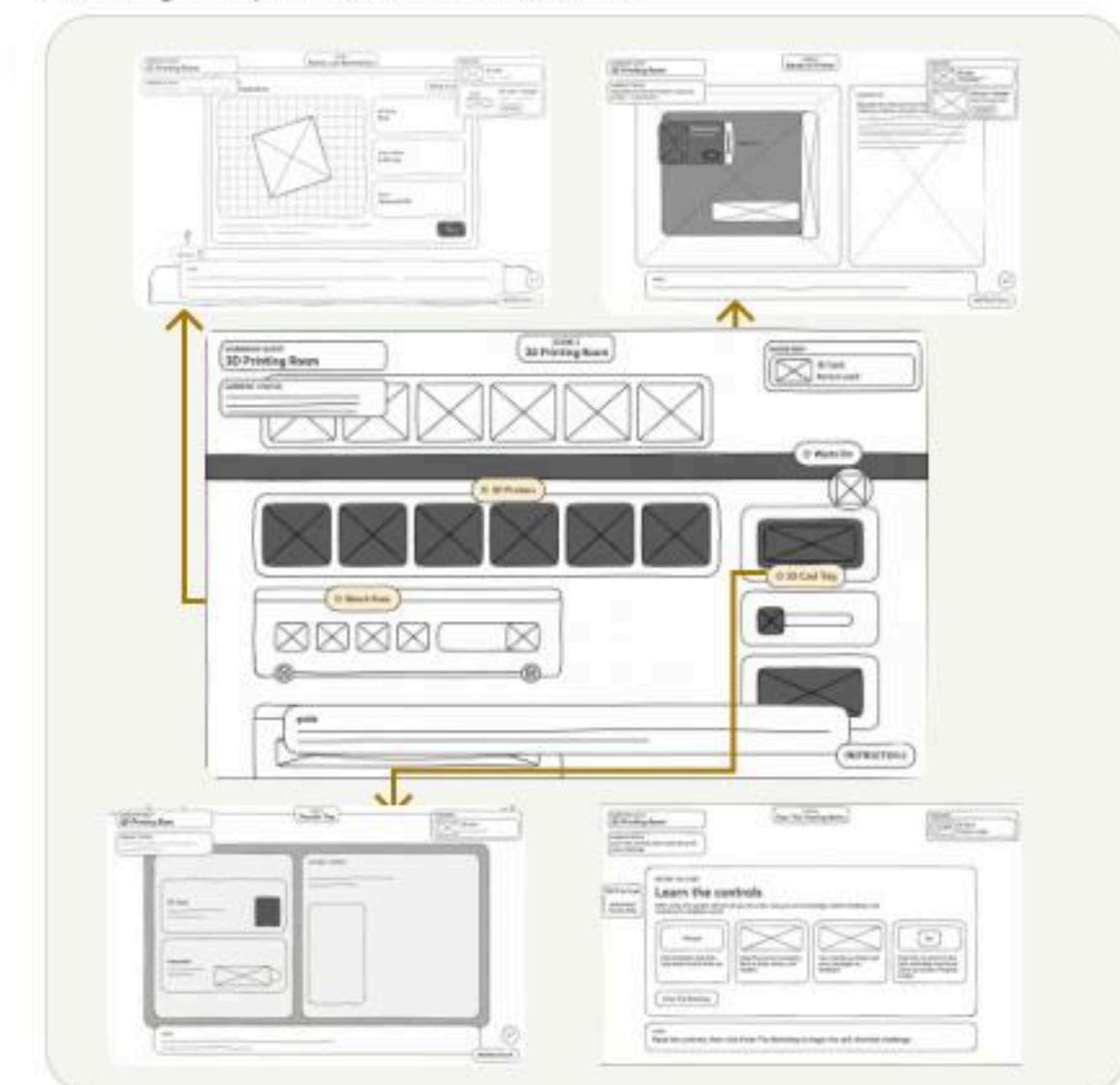
Decision: keep progression and consequences, but simplify the story layer.

SELECTED

Concept 03

3D printing simulator

A **task-based simulator** where students **practise realistic steps**: slicing, checking print time, booking, SD-card transfer, starting the printer, and collection.



What we tested

Whether **practising realistic 3D-printing steps** would build **confidence** and **retention** better than a game skin alone.

P3: I like the **interactive** aspect of it — **you're actually doing the thing**. This one actually put all what you learned into actually like doing it, which is really cool.

Realistic steps gave the **clearest confidence** and **retention signal**, making this the **strongest direction**.

Decision: take the simulator forward as the strongest bridge from game to workshop action.

Game Prototyping Journey 1

Following concept selection, the simulator was tested with the same participant profile (13 non-Design Engineering and 4 Design Engineering students) walking them through the game to surface **flaws in logic and flow**. Feedback was **clustered into categories**, with each driving a **direct design decision**, ensuring **iterations remained user-led**.

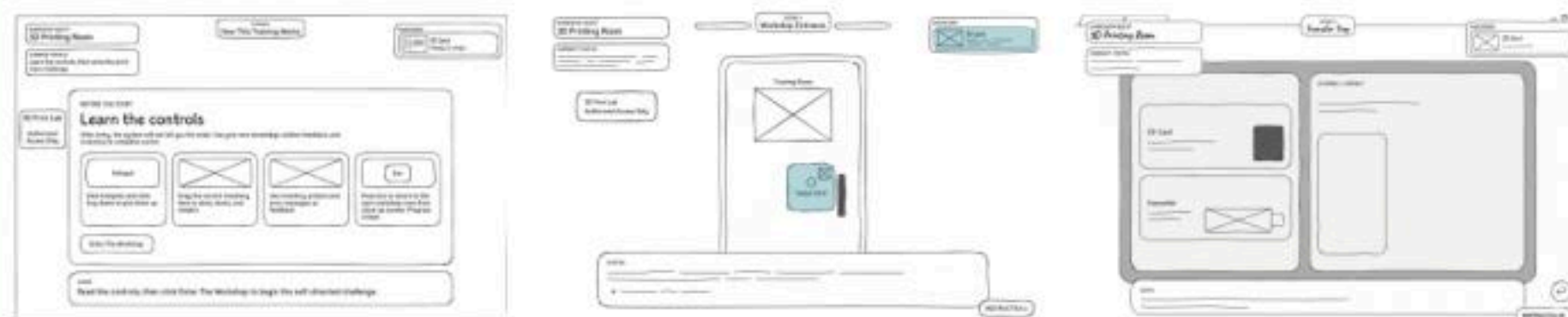
Lo-fi simulator prototyping & user testing

legend:

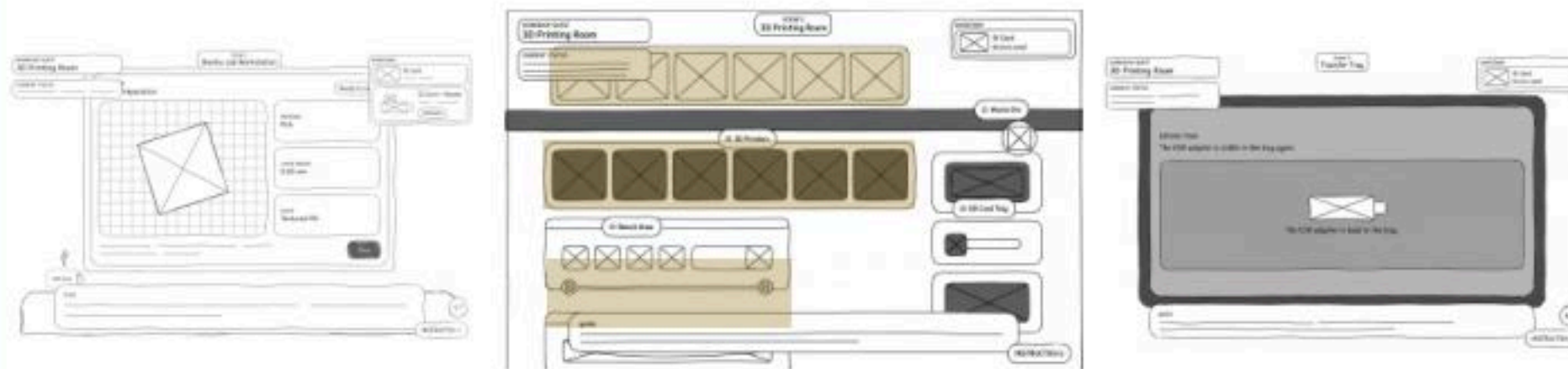
Learning & pedagogy

UX & instructions

UI & visuals

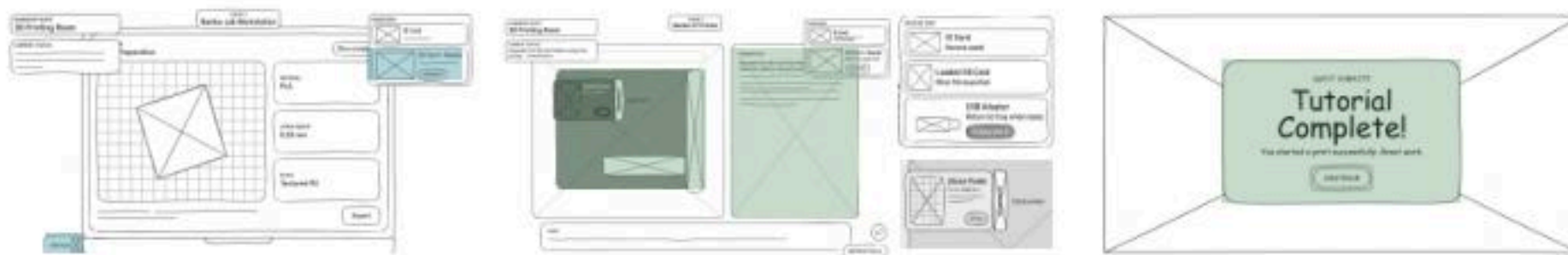


“no idea where to start, what to operate and what are the next steps.”
P14 user testing



“looks a bit **too monotonous** to me...make it **more realistic**”
P9 user testing

“let them **practice hands-on parts** ... and **learn the consequence** of actions”
P21 user testing



“How do I drag this? Click? Then plug it in ... go in... come on... go in, click, come in.”
P1 user testing

“Maybe there could be **different levels**.”
P21 user testing

Higher-fidelity simulator development

Feedback themes → learning & iteration

Learning & pedagogy

Users **wanted understanding, consequences, levels, and hands-on practice** rather than memorisation.

UX & instructions

Users were **confused** about what to **click, drag, or do next**, so the prototype needed **clearer task guidance**.

UI & visuals

Users found **hierarchy, spacing, contrast, labels, and object realism unclear** during the simulator flow.

Iterated Prototype evidence



01 Safety quiz

Simulation teaches **3D printing methodology**, while **quizzes** teach **safety knowledge**.

02 Hint system

Provided **hints** after **5 seconds** of **no movements** to guide the users when they are stuck.

03 UI authenticity

Realistic 3D printers; **interactive, rotatable** CAD model added to display support.

Outcome: User testing transformed the simulator from a **basic clickable prototype** into a **clearer, more realistic** and **guided** learning experience, adding **safety quizzes, hint support, improved visual hierarchy, and hands-on interaction** to better support induction learning.

Game Prototyping Journey 2

Final user testing: UI and UX

Methods

Participants: 14 university students

Demographics: a spectrum of **gaming exposure** (e-sport players to **single-genre players** to those with **no gaming experience**) to simulate the likely demographic range of incoming freshers.

Session: ~30 min

Users tested the **final hi-fi game prototype**.

Whenever they felt confused or unsure how to proceed, they took a **screenshot** to mark the specific UI frame. After each mark, we helped them continue. At the end, we reviewed all marked screenshots with users to identify what was confusing and what could be improved. Microsoft Clarity **heat maps** were also used to reveal **unspoken UX issues**.

Interview Quotes

UI clarity & visual hierarchy

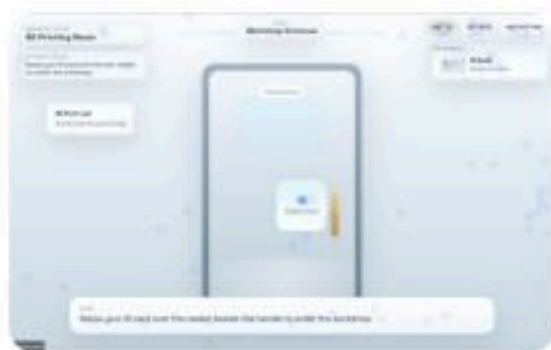
- P22: "There is a bit too much information on the page. I don't know where to look."
- P17: "Separate functional things like hint or return from the main command"
- P12: "Add dark/light or colour-blind modes, otherwise some people's eyes are just gone."

Navigation and interaction guidance

- P10: "Wait, where should I click?"
- P13: "I don't think a hint is helpful enough...it doesn't help you make progress"

Learning feedback and content depth

- P10: "Do you have explanation for why the choices go wrong?"
- P9: "What should I be expecting on the slicing screen?"



Feedback themes -> iteration

UI clarity & visual hierarchy

- Improved **visual hierarchy** so key elements like the inventory, ID card, buttons, and instructions are **easier to find**.
- Added **accessibility improvements**, including higher contrast, **colour-blind friendly colours**, clearer diagrams, and dark mode.

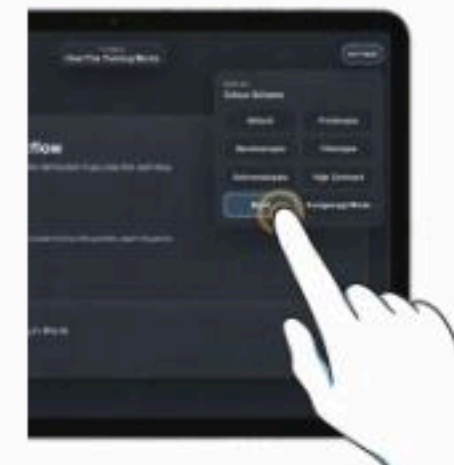
Navigation & interaction guidance

- Made the workflow more **guided** and less open-ended, so **first-time users** know what to do next.
- Added **contextual hints** when users get stuck, while avoiding over-guiding them.
- Added **dynamic** drag, drop, click hints to **mock tactile demands** in the ACE.

Learning feedback & content depth

- Added **clearer explanations** for **why actions are correct or incorrect**, instead of only saying whether the user was right.
- Added **practical 3D-printing checks** into the game, including **print time**, **dimensions**, **material use**, and **support structures**.

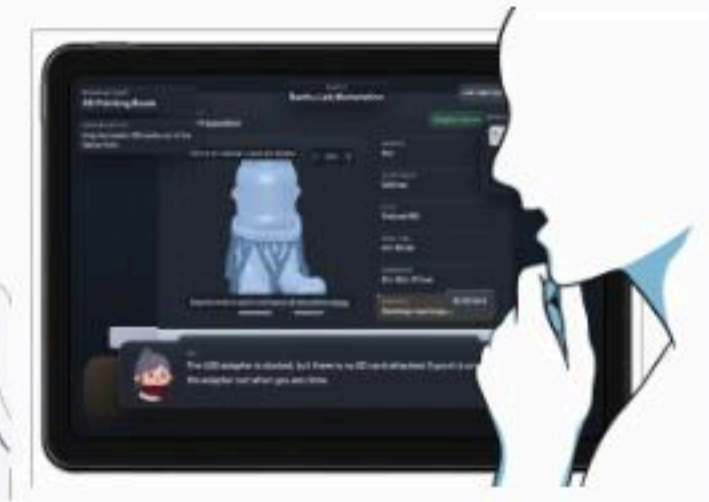
Final prototyping UX Design



- Colour modes support **different visual needs** and improve comfort during use.



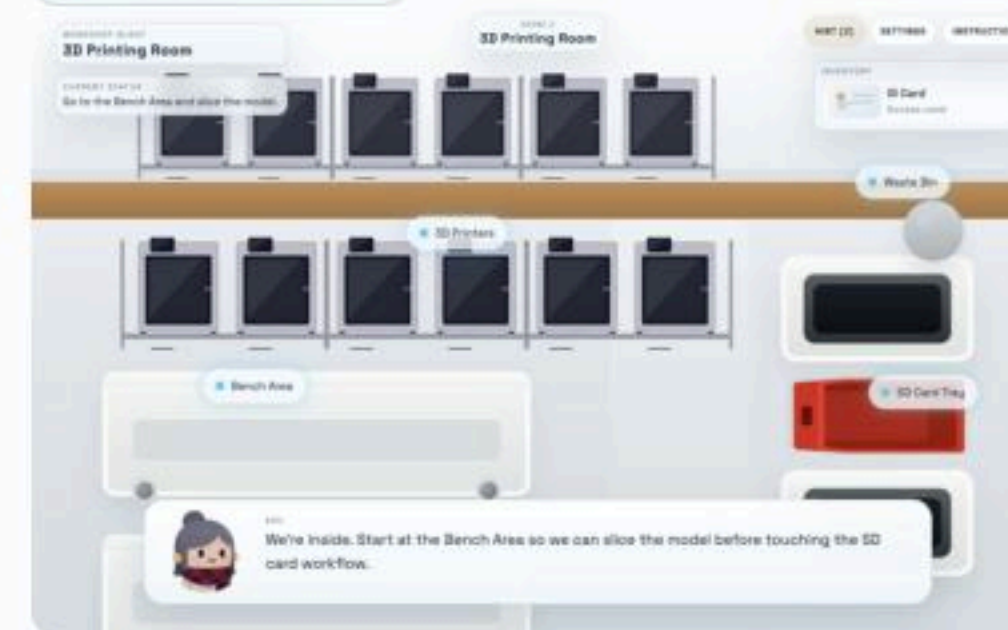
- Drag, click** and **swipe** actions are made more visible through **animation** and **visual feedback**.



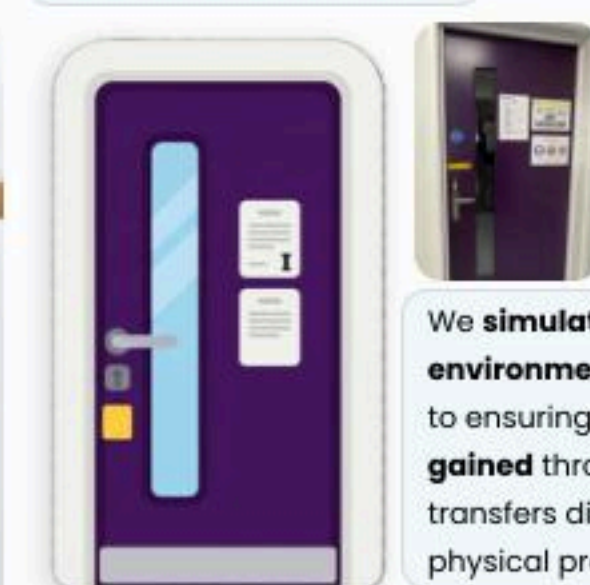
- Explanations turn mistakes into **learning moments**, improving **knowledge retention**.

UI Design

Exemplar game page



Token: ACE workshop door



We **simulated ACE environment** on our game to ensuring **knowledge gained** through play transfers directly to physical practice.

<-- 3D printing room door

In-game guide: Asu



Asu turns the system guide into a **character-led** dialogue, guiding players through the game in a more **personal** and **approachable way**. This makes the experience **feel warmer, friendlier**, and **less formal**.

Studies show that **character-like guides** can make learning feel **more engaging**, helping users **process information more deeply** than neutral system interfaces.[3]

Token: Pulsating button



Highlights the next required action, making the game flow easier to follow.

Token: Hinting system



This visually shows where an item should go, helping users understand the task **without needing direct instructions**

Reward Prototyping journey

1. Co-design: Evaluating Reward Desirability

Activity 1

Participants: 5 DE students; Session: 10–15 mins

Reward preferences

We presented **digital** and **physical** reward ideas and discussed their **desirability** with users.

Assumption 1

Digital rewards are **convenient** because they take no physical space and are easy to **customise**.

Key insight

Rejected. Users preferred **physical rewards** because they felt more **tangible, memorable**, and meaningfully connected to their **everyday life**.

“If it's just online, I would forget about it. Something physical feels more like a reward.”

Paraphrased from P8

Activity 2

Participants: 4 DE students; Session: 15 mins

Concept development

We compared **physical reward concepts** to test whether users valued **customisation** or **everyday usefulness** more.

Assumption 2

Users would prefer a **customisable** reward, as **personalisation** may make it feel more **desirable** and emotionally **meaningful**.

Key insight

Unsupported. Users saw customisation as unnecessary and preferred a reward with **clearer everyday functional use**.

“I wouldn't spend time customising it. I'd just use it as it is.”

Paraphrased from P15

2. Validation through desk research

Behavioural reinforcement in safety training

Tangible rewards at the point of rule completion drive **stronger compliance** than penalty-based or purely digital approaches. Studies show reward-based scenarios produce a **7% higher compliance rate** than penalty-based ones [4]

Across ACE's cohort of **~400 students** per year, this could mean an estimated [4]

30+ students

reliably retaining and acting on workshop safety rules.



3. Iterative design through Co-design feedback

Core design principle from user wants:

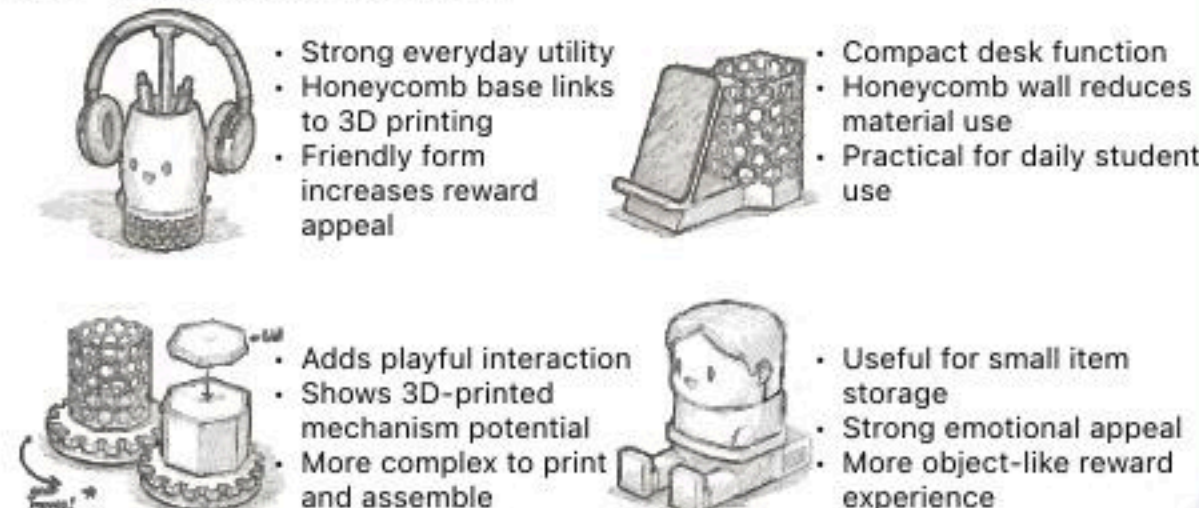
- Utility → students share the same need: **storing stationary**
- The reward reinforces the skill being learned → **3D printed**
- Keeps the object lightweight → **minimises filament usage**

Exploring Honeycomb elements

-- Reducing filament and providing lightweight



Low-fi prototype sketch



3. User testing: Desk-tidy prototype

Desk-tidy prototype testing

Demographic: 6 DE students

Desirability: "Would you actually keep this — and where?"

Practicality: "Does this fit into your daily life at ACE?"

Worth having: "Does this feel like something you earned?"



Decision: Seated character desk-tidy reward

We saw a pattern that users preferred the **seated character** because it felt useful, characterful, and more **reward-like** than the other desk-tidy options. We selected it as the final reward concept because it balanced **practicality, personality, and collectability**.

High-fi prototype



Completing the simulator unlocks the **STL file** that student can **print themselves**, giving them a **tangible outcome** that reflects their 3D-printing learning.

System Prototyping Journey

1. Simulator game limitations

A simulator cannot replicate the **physical reality** of operating machinery, meaning students may still arrive at the workshop **underprepared** for the **tactile** and **spatial demands** of hands-on fabrication.

Desk research

STEM students in **kinaesthetic learning** scored around

6%

higher on average.

Traditional **lecture** students were

1.5x

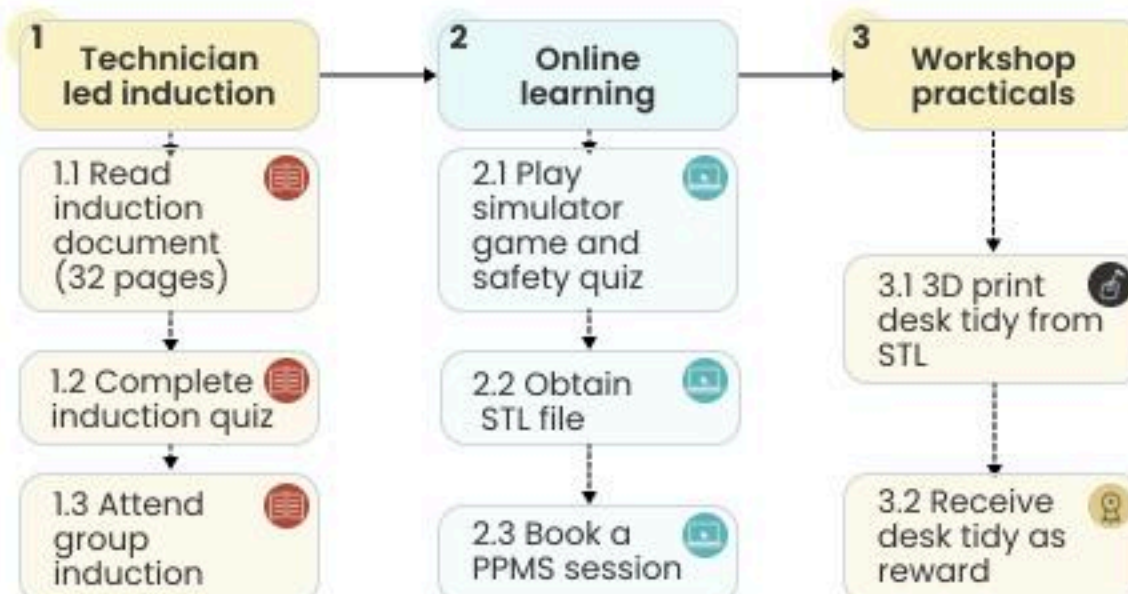
more likely to fail. [5]

Outcome: an induction **SYSTEM** with hands on practice

2. Initial system prototype

We brainstormed a **three-phase system** — **technician-led induction, online learning, and workshop practicals** — that **progressively builds student knowledge** before they touch any equipment.

Online In Ace



3. Card-sort user testing

Aim

To **evaluate** whether the three-phase onboarding structure and its content are **intuitive** and **appropriately sequenced** from the student's perspective, identifying any steps that feel **misplaced, unclear, or unnecessary** before finalising the system.

Participant Demographics & session length

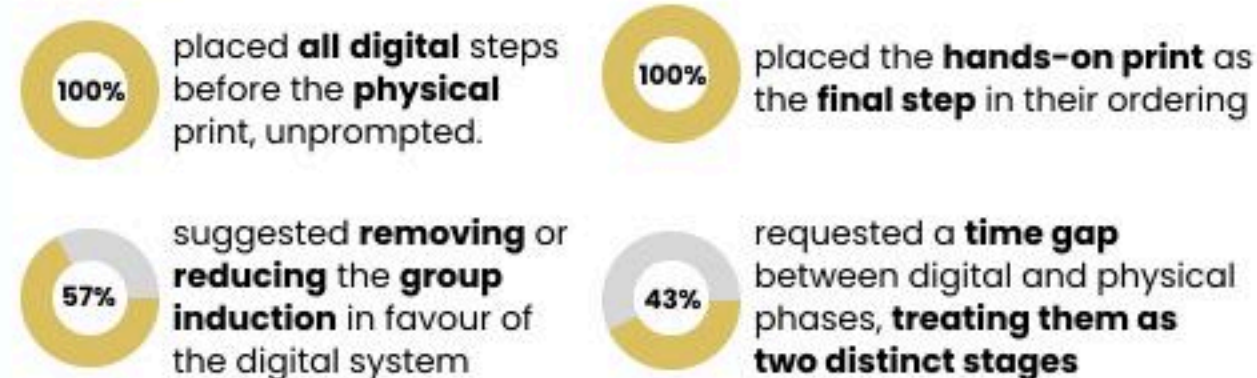
13 students: 7 non DE students and 6 DE students; 30 mins



Method

We designed our own **process cards** — each representing **one system step**, with category labels **hidden** on the reverse — so that students would sort them by **instinct** rather than our intended structure, **surfacing any gaps** between **design intent** and **user mental models**.

Results



See Google Drive: "Photographs, Recordings and Transcripts --> Photographs --> user testing final round"

4. Testing changed the system architecture

Card sorting showed users naturally grouped the induction into two phases: **digital certification** first, then **physical execution** in the workshop.

"... a **summary** would really help."
P17, user testing

1. Added a summary cheatsheet
Summarised the **procedures** of 3D printing and **safety hazards**.

"...I'd be thinking — why are there still two quizzes? That's when I'd get fed up."
P18, user testing

2. Merge quizzes
Induction and safety quizzes become one certification gateway.

"A **link** on Blackboard and then ...in the 3D printing room ... I would prefer to use a **QR code**"
P22, user testing

"You could do like the **QR code** kind of thing as well, it links in."
ACE workshop manager (Ingrid)

"For me, that would be nice if it was accessible beforehand."
ACE workshop manager (Ingrid)

3. Simulator access points
Digital link on Blackboard and **Scannable QR code** next to printer.

"I think... one week. I wouldn't want to rely on **short-term memory**. I'd want to double check."
P17, user testing

"The simulator should be completed close to the technician-led induction, ideally within the same week."
P10, user testing

"Two or three days I think. While I haven't forgotten yet."
P19, user testing

"Don't leave too long a gap. put it in the same week."
P4, user testing

Users unanimously agreed the gap should be no longer than one week.

4. Set time interval
A week is set between technician led induction and the simulator game for the users to adapt the workflow better

System validation: see Appendix - Validation (P1)



Asu's Guide to the ACE Workshop

A GAMIFIED INDUCTION FOR THE 3D PRINTING ROOM

Value proposition: A self-directed simulator that turns forgettable induction content into practisable, on-demand knowledge, giving every student an equal shot at working independently.



MEET ASU

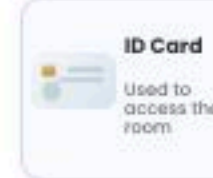


Asu is your guide through the game. She explains each step with friendly conversation and helpful hints.



YOUR INVENTORY

Key items are tracked in your inventory, making each step concrete and memorable.



ID Card
Used to access the room.



USB Reader
For SD conversion



SD Card
Ready for printing



ADDRESSING USER NEEDS

STUDENT

- ✓ Low-pressure environment builds independent confidence
- ✓ Revisit workflows anytime, no technician needed

TECHNICIAN

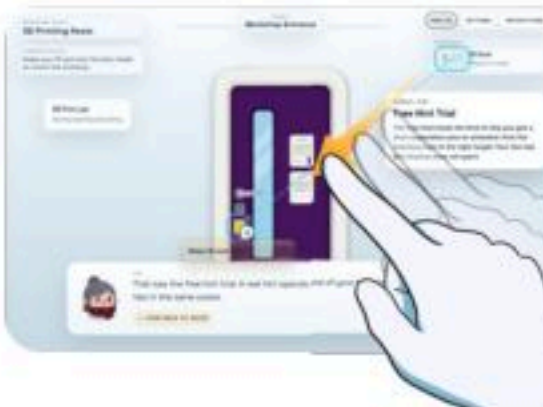
- ✓ Autonomous operation protects technician time
- ✓ Prepared students reduce peak-period interruptions

1 Introduction

Start your adventure and complete the 3D Printing induction.



Enter the room by swiping your college ID card on the door.



2 Micro SD converter

Explore the 3D printing room.

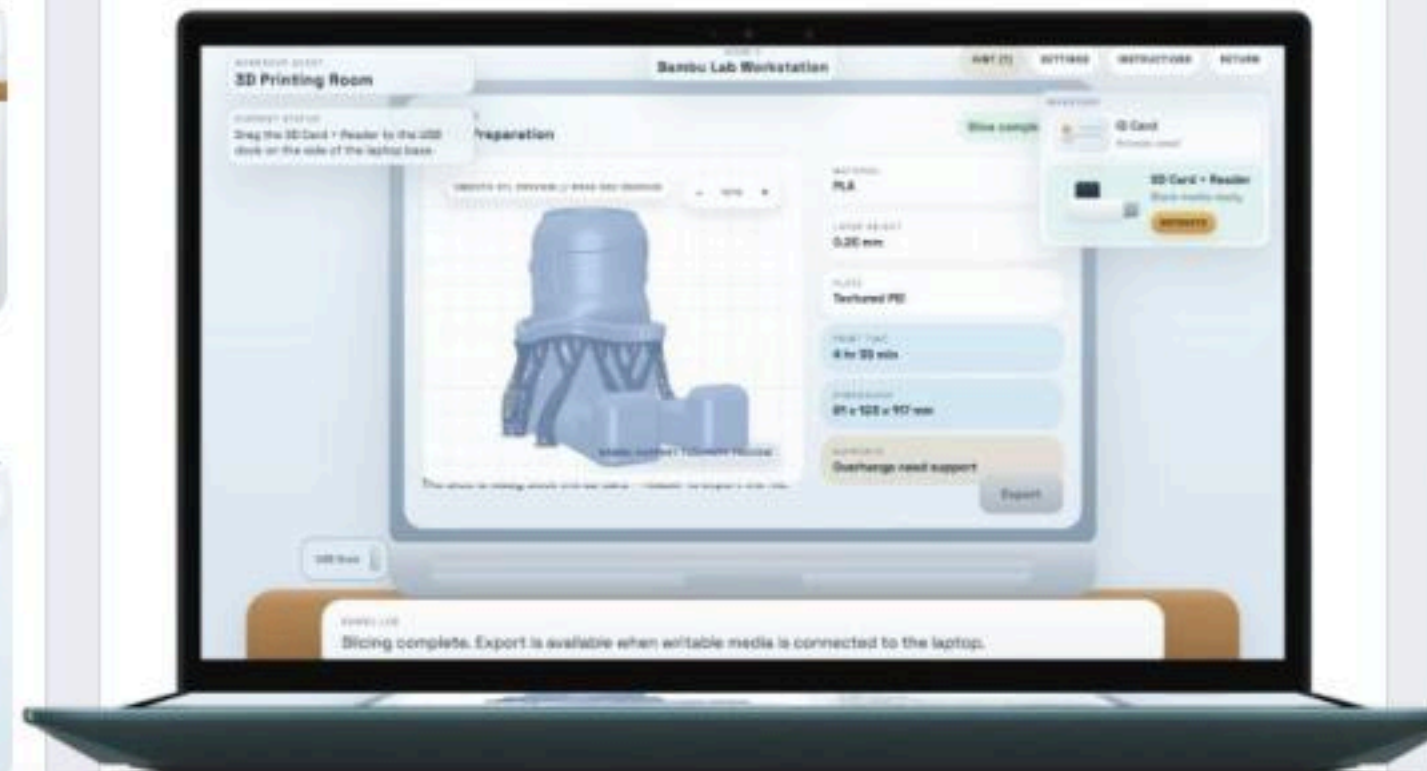


Collect the micro SD card & converter from the tray and assemble.



3 Bambu Lab Workstation

Inspect and prepare your 3D model for printing. Choose **material**, **layer height** and **supports**. Check **model size**, **print time** and ensure everything is ready before **exporting**.



4 Time to print

Start your print after inserting the SD card into the right slot.



Finally, **maintain good lab practice** and return the converter to the tray



5 Hinting system

Utilise the hinting system to help you move along.

MANUAL HINT

Next Step

Drag the loaded USB reader left, away from the laptop dock. It will return to inventory automatically.

The system will demonstrate where to **drag/drop your item** when hints are used, or if it detects that you understand the action but have slight issues with game controls.

HINT (2)

Drag Item here

A limited number of 2 hints ensure that students can get a **moderate amount of guidance** while also having to exhibit critical thinking.

THE COMPLETE INDUCTION SYSTEM WORKFLOW



COMPLETE THE JOURNEY. GET YOUR REWARD!

Everyone who completes the digital learning phase, plays the game, and finishes the workshop induction will receive their own personal ASU DeskTidy!



Your achievement.
Your DeskTidy.
Your ACE Workshop journey.

Appendix - Validation: Mixed-Method Validation

Validation aim

We tested whether the simulator tutorial moved users beyond passive reading by improving recall of key induction knowledge and the 3D printing workflow.

Validation method

Quantitative: 10-question pre / post test

Qualitative: requirement rating

1. Validation method

Quantitative Validation

1. Read the induction document
2. Pre-test
3. Simulator game
4. Post-test
5. Calculate learning gain

Participants

- **Quantity:** 20
- **Demographic:** Students with diverse range of 3D printing experience
- **Duration:** 45 min

Qualitative Validation

We mapped the final concept against our defined requirements to qualitatively validate how well the solution addressed the original problem space.

2. Quantitative validation

Sample multiple choice question sets

Pre-test	Post-test
What does clicking Slice in Bambu Lab do?	In the laptop software, what is the purpose of slicing the model?
A desk tidy needs to be functional while staying within workshop material limits. Which slicing decision is MOST appropriate?	When preparing a functional print under resource constraints, what is generally considered good practice?
What print setting most directly changes how visible the layer lines are?	Which setting has the biggest impact on how smooth the printed surface looks?
Which parts of an FDM 3D printer are most likely to be dangerously hot?	Which parts of an FDM 3D printer are most likely to cause burns if touched during or shortly after printing?

Quantitative Statistics

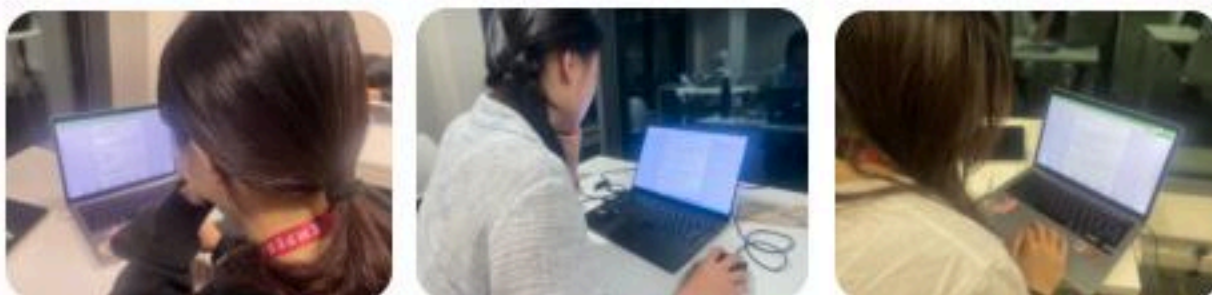
4.8 / 10 Average Pre-test

7.1 / 10 Average Post-test

90% (18/20) Participants Improved

+2.3 Average Learning Gain

Learning Gain equals to:
 $\text{post-test score} - \text{pre-test score}$
showing an average of how much users improved after using the simulator game.



3. Qualitative validation

We rated the final concept against success criteria to evaluate how well it addressed the original problem space.

Criteria

(same as page 2)

CR1 Effective learning

CR2 On-Demand Access

CR3 Independent Confidence

CR4 Engagement

CR5 Induction complement

Success requirements

Generated from requirements

Post-game score measurably higher than pre-game baseline across ≥70% of users.

Content is accessible 24/7 at any locations

Game completes end-to-end without any technician input across 100% of sessions

How strongly it motivates users to complete the induction.

Students confirm the game reinforces rather than replaces induction

The five success criteria were plotted on a radar chart to provide an at-a-glance visual assessment of how evenly the final concept performs across all defined requirements.

Benchmarking maps the traditional induction against the simulator game across all five requirements, revealing where the game meaningfully advances on the current system and where gaps remain.



4. Validating System Feasibility in ACE

ACE workshop manager: Ingrid

"It can be used as a basic introduction into 3D printing, but it can also be used as a refresher — something that could be easily made accessible that students can access at any point, rather than just once a year."

"If every student were to print something small as part of their induction, would that be feasible?"

"The amount of waste that we see from failed prints... we probably spend more money on that than we do on the induction side of it."

"It might actually help reduce the waste that the students produce — if they're a lot more aware of ... the kind of mistakes to avoid."

GBP 6,485.15 spent

8.6 kg failed waste



5. Limitations & next steps

Limitations

- Vibe-coded due to complexity of game development; token limits reduced UI fidelity and left one unresolved bug in the looks-like prototype
- Cannot be made compulsory without adding pressure to mandatory induction components

Next steps

- Develop formal game development skills to raise technical fidelity
- Introduce difficulty levels tailored to varying 3D printing experience
- Expand machinery coverage to include laser cutter and beyond

Appendix – Bibliography and Product Links

Bibliography

1. Imperial College London (no date) 'ACE workshop', Faculty of Engineering. Available at: <https://www.imperial.ac.uk/engineering/departments/design-engineering/about-us/design-engineering-spaces/ace-workshop/> (Accessed: 2 June 2026).
2. Legaki, N.-Z., Xi, N., Hamari, J., Karpouzis, K. and Assimakopoulos, V. (2020). The effect of challenge-based gamification on learning: An experiment in the context of statistics education. *International Journal of Human-Computer Studies*, [online] 144, p.102496. doi:<https://doi.org/10.1016/j.ijhcs.2020.102496>.
3. Domagk, S. (2010). Do Pedagogical Agents Facilitate Learner Motivation and Learning Outcomes? *Journal of Media Psychology*, 22(2), pp.84–97. doi:<https://doi.org/10.1027/1864-1105/a000011>.
4. Lee, B. and Kim, H. (2024) 'Evaluating the effects of safety incentives on worker safety behavior control through image-based activity classification', *Frontiers in Public Health*, 12. doi: 10.3389/fpubh.2024.1430697.
5. Freeman, S., Eddy, S.L., McDonough, M., Smith, M.K., Okoroafor, N., Jordt, H. and Wenderoth, M.P. (2014) 'Active learning increases student performance in science, engineering, and mathematics', *Proceedings of the National Academy of Sciences*, 111(23), pp. 8410–8415. doi: 10.1073/pnas.1319030111.
6. Designer Xueli (2025) 'Making games with Figma AI is surprisingly easy', Xiaohongshu, 10 October. Available at: <https://www.xiaohongshu.com/discovery/item/68e85c7d0000000005013cd5> (Accessed: 2 June 2026).
7. Service Design Weave (2022) 'Service design methods: service prototyping', Xiaohongshu, 18 December. Available at: <https://www.xiaohongshu.com/discovery/item/639f24a2000000001f012540> (Accessed: 2 June 2026).
8. KOBETruth Art and Design Education (2024) 'Useful service design tool: what is desktop walkthrough?', Xiaohongshu, 6 December. Available at: <https://www.xiaohongshu.com/discovery/item/675283820000000007038c76> (Accessed: 2 June 2026).
9. MOL (2025) 'RCA service design portfolio 3', Xiaohongshu, 8 August. Available at: <https://www.xiaohongshu.com/discovery/item/6895999900000000022021678> (Accessed: 2 June 2026).
10. AiSi Interaction Design Studio (2025) 'A hidden gem for interaction designers: paper prototyping', Xiaohongshu, 31 July. Available at: <https://www.xiaohongshu.com/discovery/item/688b639d000000000303258e> (Accessed: 2 June 2026).

Important links

1. Google Drive – Project Archive

This shared Google Drive acted as our central project archive, storing key materials across design, feedback, inspirations, planning, activities, prototypes, recordings, handwritten notes and transcripts. It helped the team keep research evidence, development files and portfolio resources organised in one place.

https://drive.google.com/drive/folders/1odAb2QI7hp-iDbAhNlfJMgyWrLvphGNe?usp=share_link

2. Microsoft Loop – Meeting Log

This Microsoft Loop page documents our 12 team meetings held within less than one month. Each log includes the agenda, attendees, key findings, task allocation, to-do list and meeting summary, helping us track decisions, progress and individual responsibilities across research, prototyping and portfolio development.

All meeting logs could be found on the Google Drive link above, inside the folder of “Meeting Logs”.

3. Final Simulator Game (Check Github Page)

<https://diethyletherene.github.io/Asu-s-guide-to-the-ACE-workshop/>

Appendix - User Testing

Users involved in our project

P1: Jiashuo
P2: Joseph Birch
P3: Liam
P4: Yubo
P5: Adele
P6: Nancy
P7: Sherwin
P8: Joyce
P9: Cris
P10: Shana
P11: Tim
P12: Kang
P13: Tony
P14: Grace
P15: Brian
P16: Sohaib
P17: Kerry
P18: Sihan
P19: Aiden
P20: Jerry
P21: Ananya
P22: Charlotte
P23: Noah
P24: Ikem
P25: Gautham

Validation results

Participant	Pre-test /10	Post-test /10	Learning Gain
P1	4	7	+3
P2	5	7	+2
P3	3	7	+4
P4	6	8	+2
P5	4	6	+2
P6	5	8	+3
P7	7	8	+1
P8	6	6	0
P9	4	7	+3
P10	5	8	+3
P11	3	6	+3
P12	6	9	+3
P13	5	4	-1
P14	4	8	+4
P15	7	9	+2
P16	5	8	+3
P17	2	5	+3
P18	6	8	+2
P19	4	5	+1
P20	5	8	+3

Learning gain distribution

Gain	Participants
+4	2
+3	9
+2	5
+1	2
0	1
-1	1

Consent form

The consent forms are titled 'Participant Consent Form' and include the following conditions (checkboxes are marked with checkmarks):

- I consent to take part in the above study conducted by the student of [University Name] as part of their academic activities at [University Name].
- I understand that the information I provide will be used for research purposes only.
- I understand that I can withdraw at any time without giving any reason and without my legal rights being affected.
- I understand that the information will be recorded using a recording device. I understand that the recording will be stored on a secure server and will be accessible only to the research team.
- I understand that the information will be used in the research report and will be made available to the public.
- I understand that the information will be used in the research report and will be made available to the public.
- I understand that the information will be used in the research report and will be made available to the public.

Each form also includes a space for the participant's signature and date.

User testing photos



Appendix - Supporting Visuals

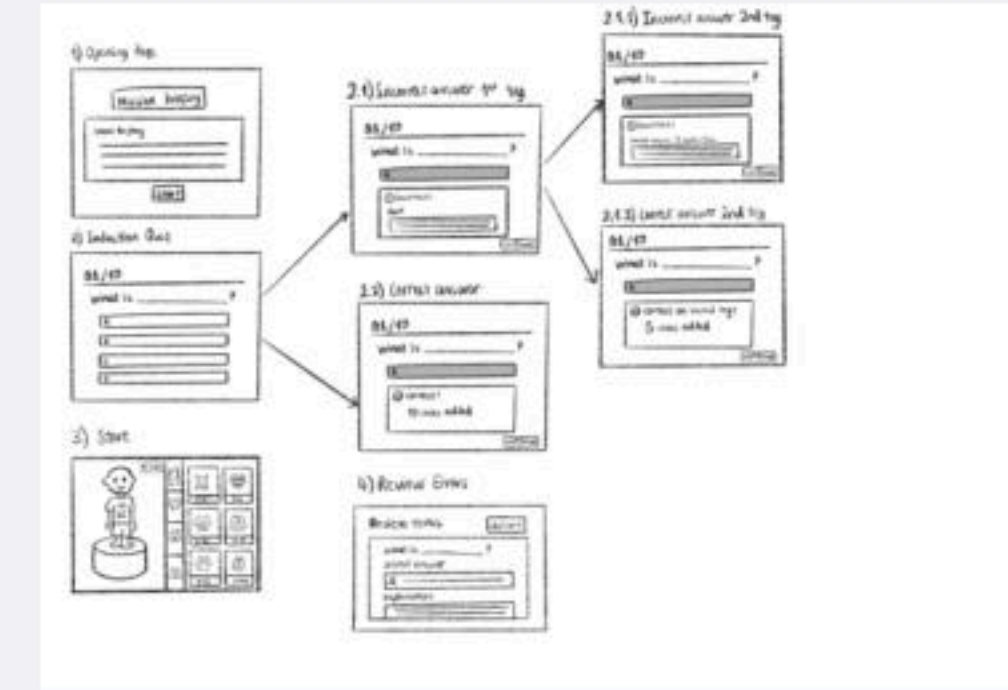
App icon



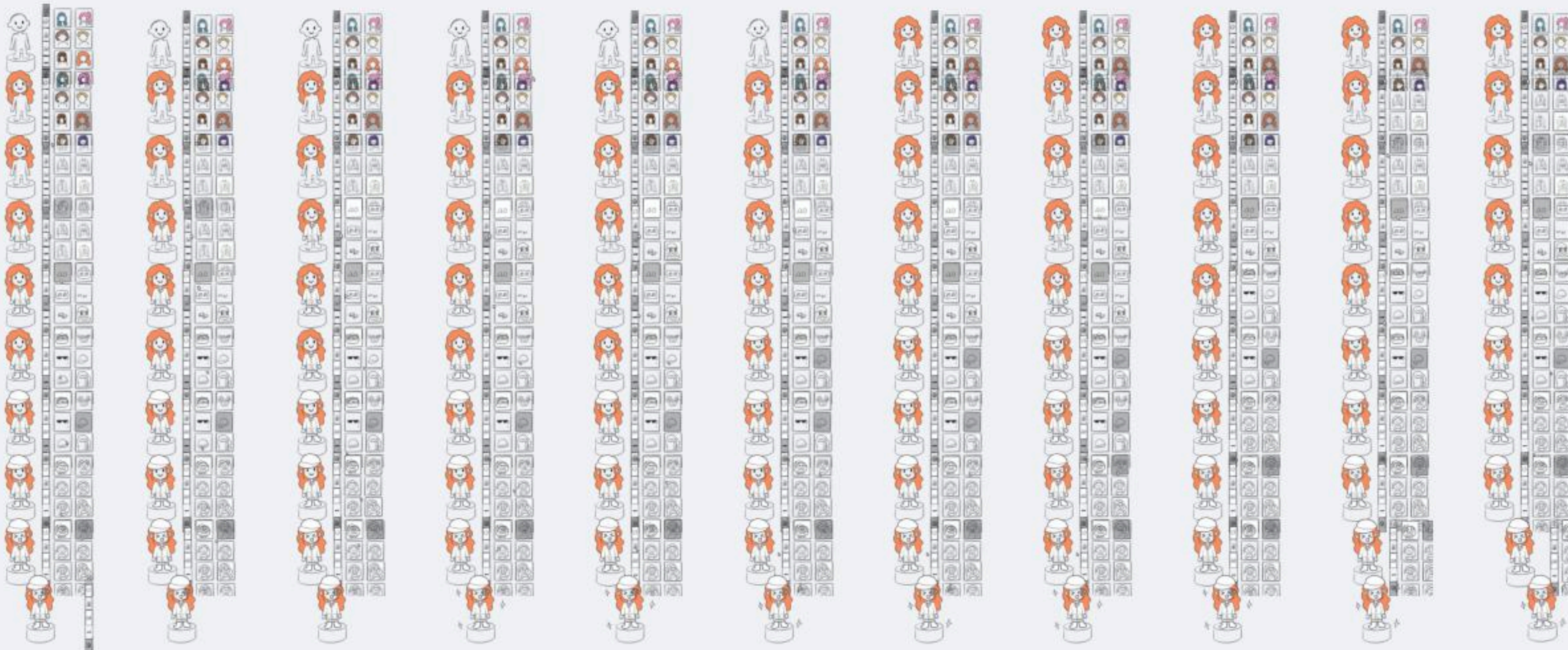
Storyboard Development



Wireframe Development



Dress-up quiz game animation development



Appendix - AI Statement

AI was used selectively to condense written content for layout purposes and to assist with game coding due to technical constraints. All research, user testing, synthesis, design decisions, and creative direction remained entirely our own.